

Sugandhar K

AI Engineer · Computer Vision Engineer · LLM & GenAI Engineer · MLOps

ksugandhar123@gmail.com | +91 93610 51919 | linkedin.com/in/sugandhark | github.com/Sugandharkannan | Chennai, India

PROFESSIONAL SUMMARY

AI Engineer with **2+ years** of production experience in **Computer Vision** and **Generative AI**. Shipped 3 end-to-end production systems: **YOLOv8 PPE monitoring** (90%+ mAP, 25+ FPS on live CCTV), **multi-camera ANPR** with DeepSORT tracking, and an **enterprise RAG assistant** serving sub-second responses over 1,000+ docs. Optimised inference **12x** via threaded pipelines; skilled in **LLM fine-tuning** (LoRA/QLoRA/PEFT), **RAG** (LangChain + FAISS), and **FastAPI + Docker** deployment.

TECHNICAL SKILLS

Languages	Python (Advanced), SQL, Bash/Linux, Git/GitHub
CV & Vision	YOLOv8, YOLOv5, OpenCV, Object Detection, PPE Monitoring, Video Analytics, RTSP, ONNX, TensorRT, Edge Deployment, DeepSORT, Real-Time Inference Optimisation
ML / DL	PyTorch, TensorFlow, Scikit-learn, Transfer Learning, Model Quantisation, Hugging Face Transformers
GenAI / LLMs	LLaMA 3, Mistral, Qwen2-VL, LangChain, LangGraph, RAG, FAISS, ChromaDB, Qdrant, LoRA/QLoRA/PEFT, Prompt Engineering, Vision-Language Models (VLM), Agentic AI
MLOps & Infra	FastAPI, Docker, Kubernetes (pod/deploy/service/ingress), GitHub Actions CI/CD, AWS (EC2, S3, ECR), MLflow, Weights & Biases, Triton Inference Server

PROFESSIONAL EXPERIENCE

SeeWise AI – LMBees Pvt. Ltd Chennai, India
Applied Data Scientist – Computer Vision & GenAI Dec 2024 – Present

- PPE Safety:** YOLOv8 on multi-camera CCTV — **90%+ mAP, 6 classes, 25+ FPS**; alerting pipeline eliminated **60%** of manual monitoring overhead (validated vs. pre-deployment guard hours)
- ANPR System:** Multi-camera tracking (YOLOv8 + DeepSORT + EasyOCR) at **30 FPS** with cross-camera entry/exit logging; reduced monitoring effort by **80%**
- RAG Assistant:** LangChain + LLaMA 3 + FAISS over **1,000+ enterprise docs**; citation-backed answers via FastAPI in **<1 s**; multi-turn memory support

SeeWise AI – LMBees Pvt. Ltd Chennai, India
Machine Learning Associate – Computer Vision Mar 2024 – Dec 2024

- Trained **YOLOv8/YOLOv5** to **88–92% mAP**; exported to **ONNX** and optimised via **TensorRT** for edge deployment on industrial hardware
- Improved dataset quality by **40%** using Roboflow and Label Studio; reduced false-positive rate **35%** via NMS tuning and class-weighted training

Sweng Solutions Remote
Machine Learning Intern – NLP & AI Mar 2023 – Feb 2024

- Built NLP semantic search chatbot (FAISS + sentence-transformers); achieved **35% relevance improvement** over TF-IDF baseline; reduced query latency **40%** via async batching and INT8 quantisation

KEY PROJECTS

Industrial Safety Monitoring System [YOLOv8 · OpenCV · TensorRT · FastAPI · Docker]

PPE detection at **90%+ mAP / 25+ FPS**; **12x throughput uplift** via TensorRT FP16 + ThreadPoolExecutor; alerting dashboard cut manual monitoring by **60%**.

Intelligent Vehicle ANPR & Tracking [YOLOv8 · DeepSORT · EasyOCR · FastAPI · Redis]

Multi-camera ANPR at **30 FPS**; cross-camera entry/exit state with fuzzy plate matching; reduced manual monitoring by **80%**.

Enterprise RAG Knowledge Assistant [LangChain · LLaMA 3 · FAISS · ChromaDB · FastAPI]

RAG over **1,000+ enterprise docs**; 512-token chunking, sentence-transformer embeddings, multi-turn memory; answers in **<1 s**.

Mistral-7B Fine-tuning with QLoRA [QLoRA · PEFT · Hugging Face · W&B · BitsAndBytes]

Fine-tuned **Mistral-7B** with **4-bit QLoRA** ($r=8, \alpha=16$); gradient checkpointing + accumulation for Colab T4; full experiment tracking with W&B.

EDUCATION

Anna University Chennai, India
B.E. – Electronics & Communication Engineering 2019 – 2023

CERTIFICATIONS & COURSES

Deep Learning Specialisation — *Stanford University*

Neural Networks, CNNs, Sequence Models, Hyperparameter Tuning, Optimisation